# Markowitz

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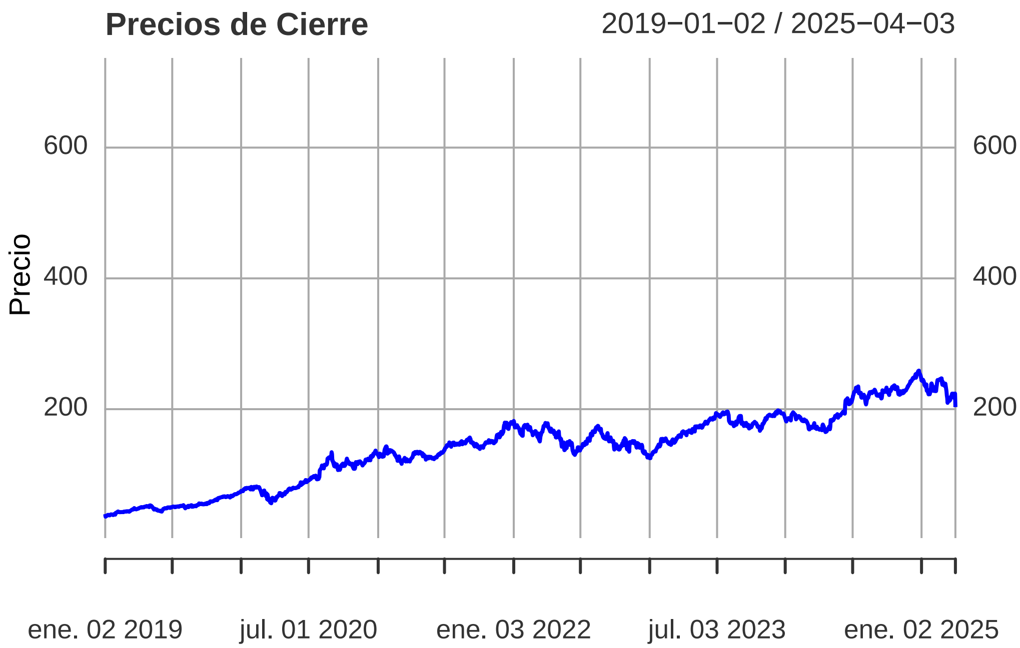
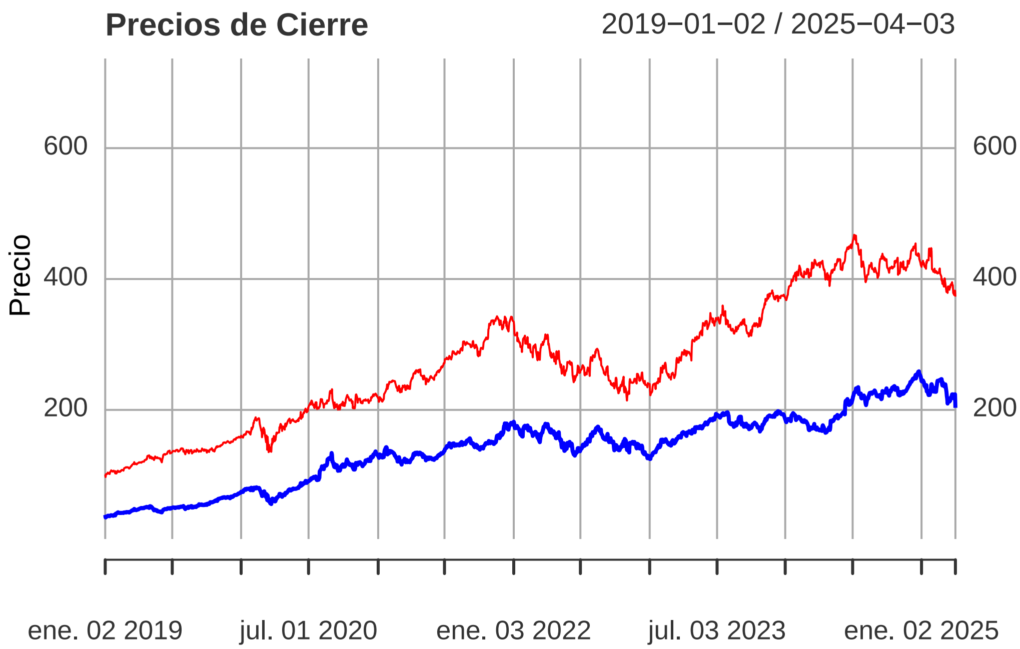
```
library(quantmod)
library(ggplot2)
library(ggfortify)
library(tseries)

precios<-c("AAPL", "MSFT", "NVDA", "AMZN", "META")

getSymbols(precios, from="2019-01-02", periodicity ="daily")
```

```
[1] "AAPL" "MSFT" "NVDA" "AMZN" "META"
```

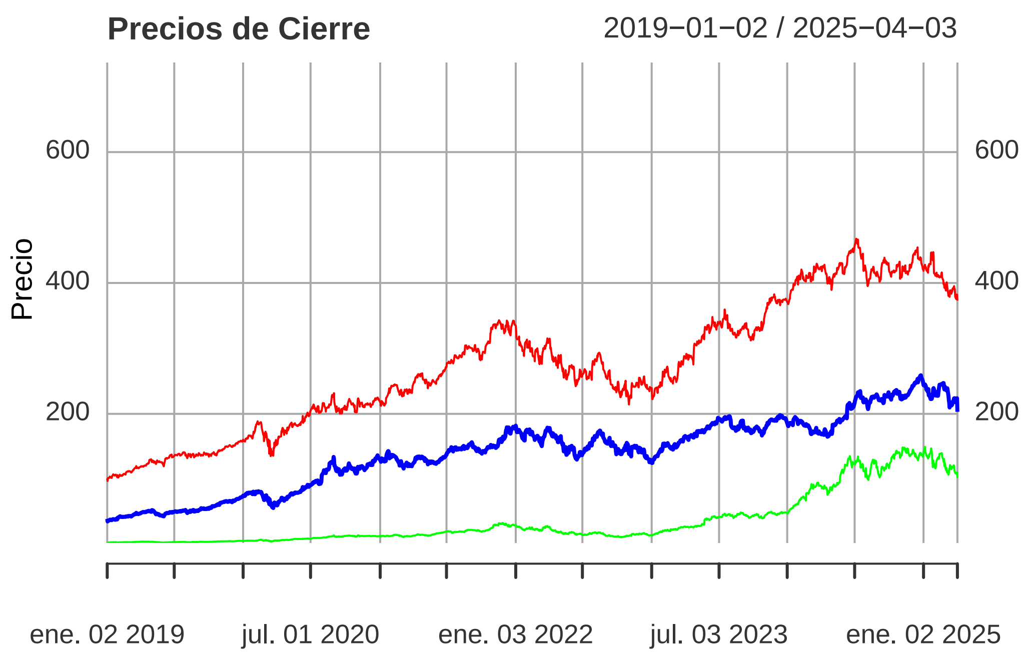
```
min_len <- min(
  nrow(AAPL), nrow(MSFT), nrow(NVDA), nrow(AMZN), nrow(META)
)

aapl <- AAPL$AAPL.Close[1:min_len]
msft <- MSFT$MSFT.Close[1:min_len]
nvda <- NVDA$NVDA.Close[1:min_len]
amzn <- AMZN$AMZN.Close[1:min_len]
meta <- META$META.Close[1:min_len]

# Graficamos solo una vez y luego agregamos líneas y leyenda
plot(aapl, type = "l", col = "blue",
     ylim = range(c(aapl, msft, nvda, amzn, meta)),
     main = "Precios de Cierre", ylab = "Precio", xlab = "Tiempo")
```



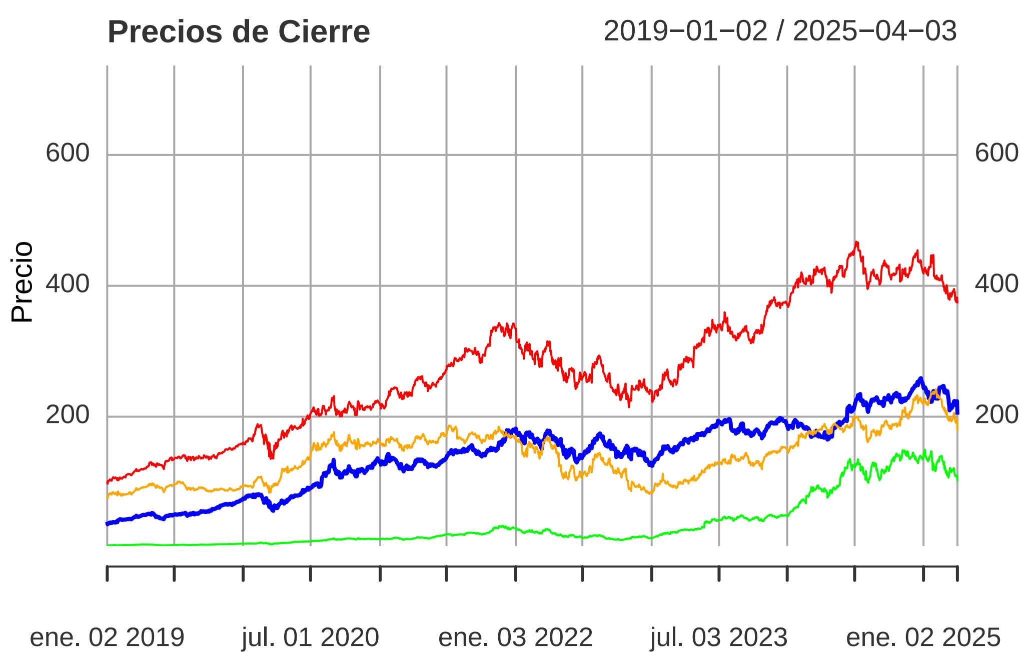

```r
lines(msft, col = "red")
```



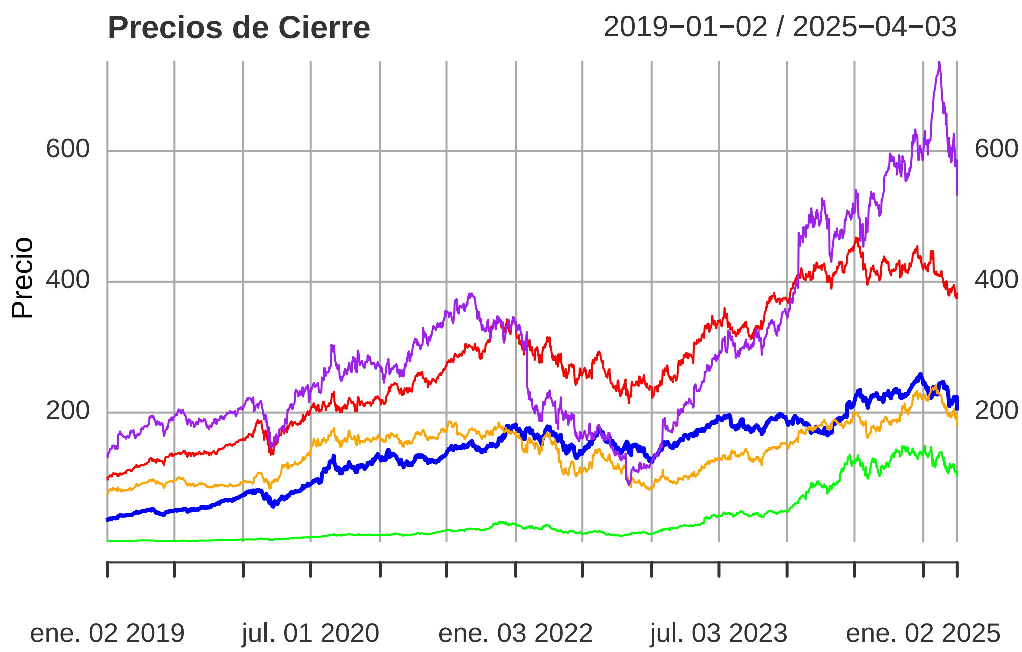
```r
lines(nvda, col = "green")
```




```r
lines(amzn, col = "orange")
```



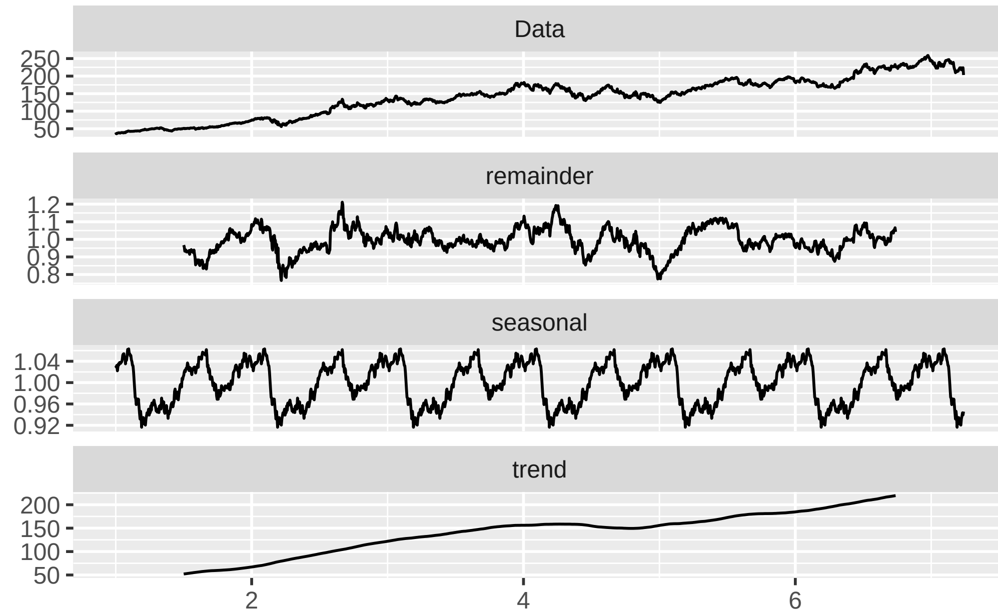
```r
lines(meta, col = "purple")

legend("topright",
       legend = c("AAPL", "MSFT", "NVDA", "AMZN", "META"),
       col = c("blue", "red", "green", "orange", "purple"),
       lty = 1, lwd = 2)
```



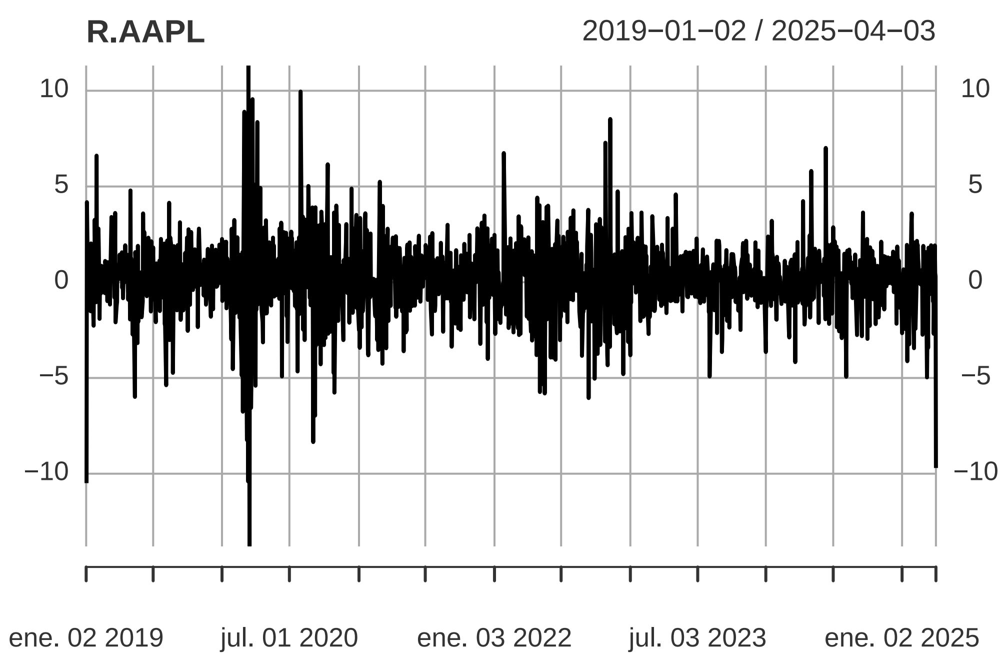

**Descomposición de los retornos porcentuales**

```r
precio_cierre<- AAPL$AAPL.Close

serie_de_tiempo<-ts(precio_cierre, frequency = 252)

descom<- decompose(serie_de_tiempo, type="multiplicative")

autoplot(descom)
```

```
Warning: Removed 504 rows containing missing values or values outside the scale range
(`geom_line()`).
```

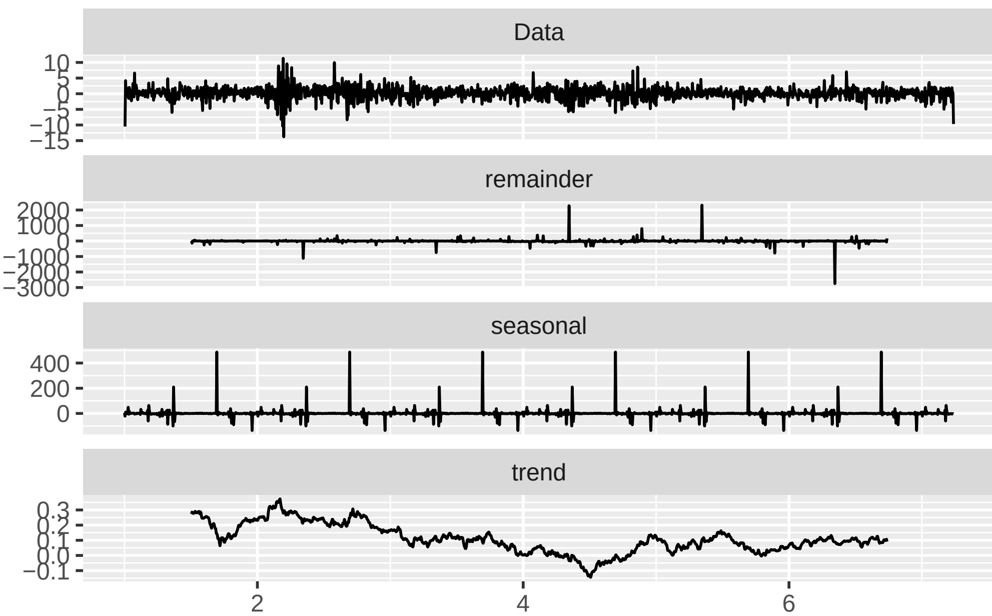



```r
R.AAPL<-diff(log(AAPL$AAPL.Close))*100

plot(R.AAPL)
```



```
serie_de_tiempo<- ts(R.AAPL, frequency = 252)

descom<-decompose(serie_de_tiempo, type="multiplicative")

autoplot(descom)
```

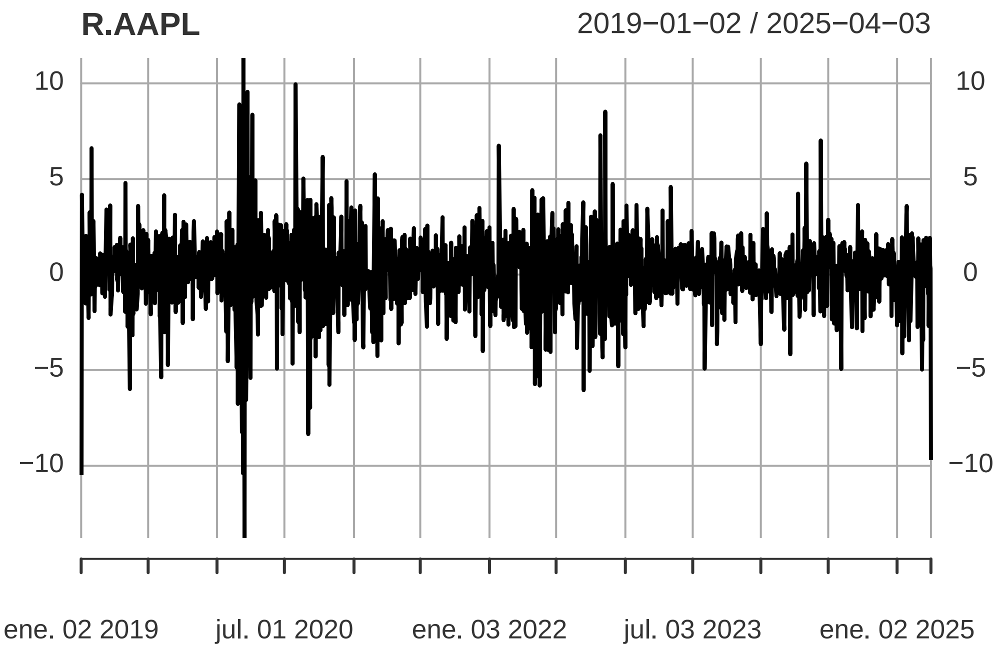
```
Warning: Removed 507 rows containing missing values or values outside the scale range
(`geom_line()`).
```



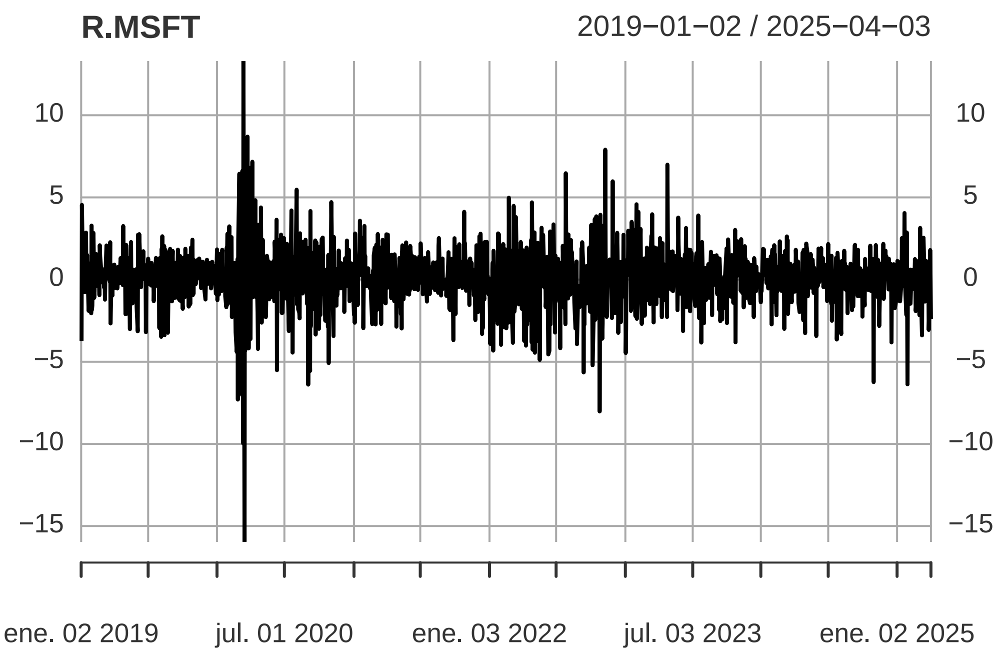

### RETORNOS DE LOS ACTIVOS

```
R.AAPL<-diff(log(AAPL$AAPL.Close))*100
R.MSFT<-diff(log(MSFT$MSFT.Close))*100
R.NVDA<-diff(log(NVDA$NVDA.Close))*100
R.AMZN<-diff(log(AMZN$AMZN.Close))*100
R.META<-diff(log(META$META.Close))*100
```

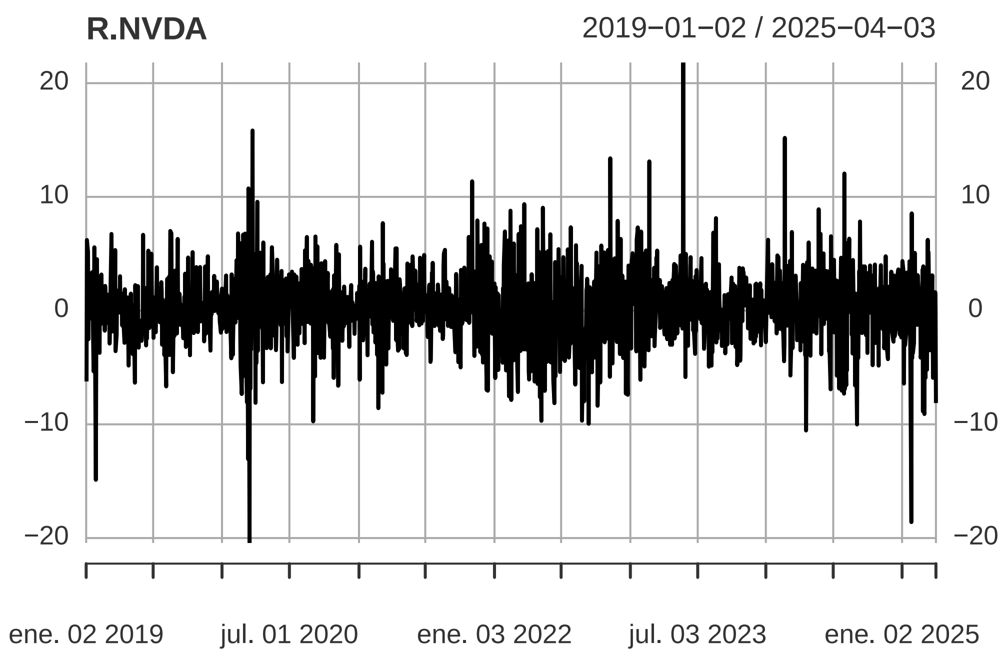
```
plot(R.AAPL)
```


R.AAPL
2019−01−02 / 2025−04−03
10
5
0
−5
−10
ene. 02 2019
jul. 01 2020
ene. 03 2022
jul. 03 2023
ene. 02 2025

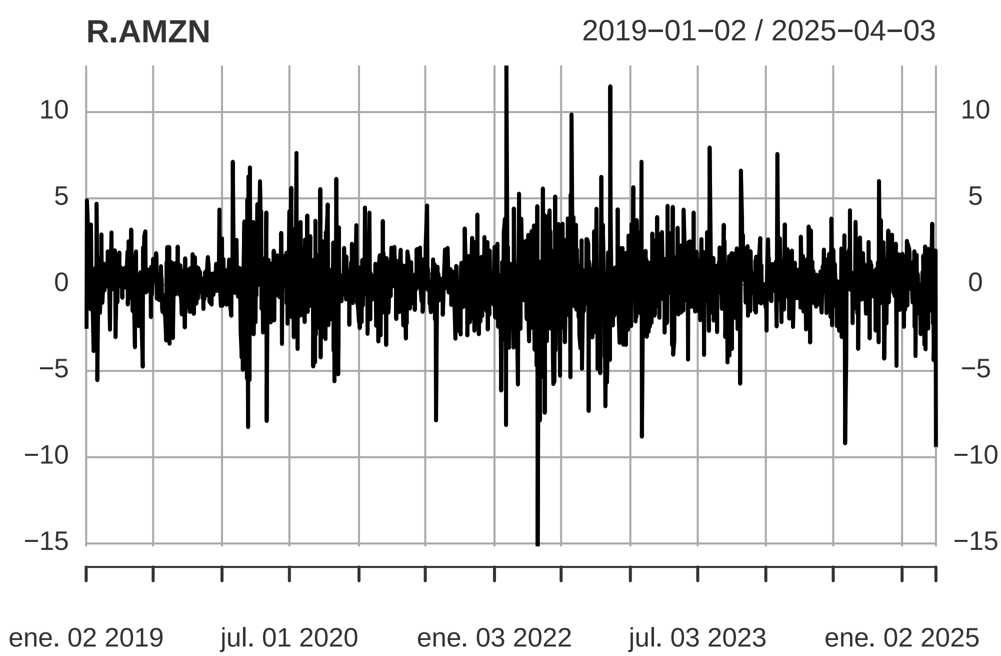
```r
plot(R.MSFT)
```


R.MSFT
2019−01−02 / 2025−04−03
10
5
0
−5
−10
−15
ene. 02 2019
jul. 01 2020
ene. 03 2022
jul. 03 2023
ene. 02 2025

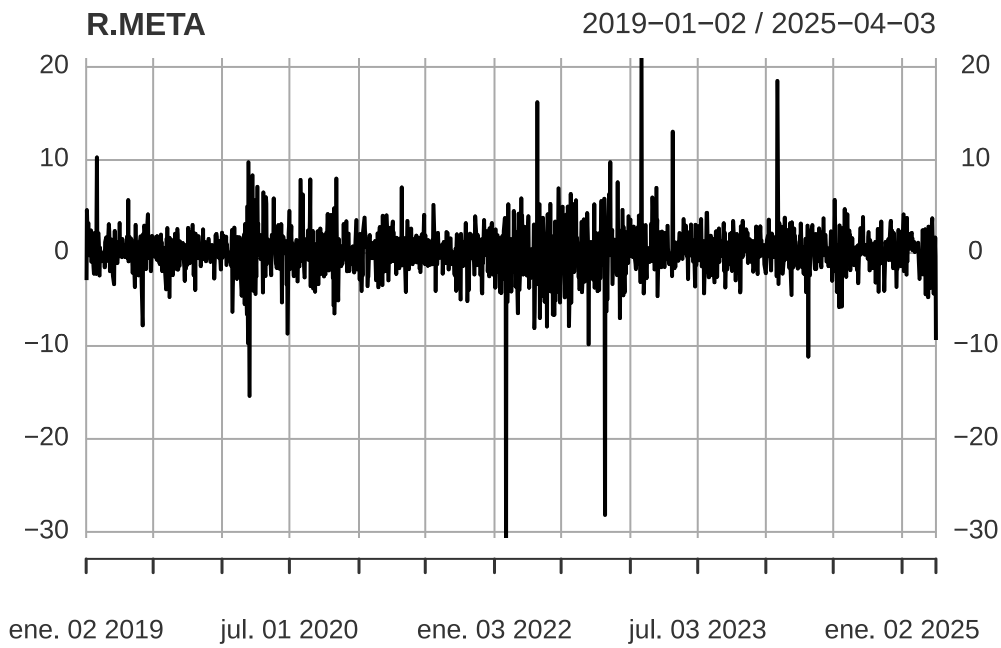
```r
plot(R.NVDA)
```




```r
plot(R.AMZN)
```



```
plot(R.META)
```




```
#VECTOR DE RETORNOS
mu<-rbind(mean(R.AAPL[-1]), mean(R.MSFT[-1]), mean(R.NVDA[-1]), mean(R.AMZN[-1]) , mean(R.ME
```

```
#datos de cierre
Dat<-data.frame(R.AAPL[-1], R.MSFT[-1], R.NVDA[-1], R.AMZN[-1], (R.META[-1]))
#matriz de varianza covarianza
s<-cov(Dat)
#vector de unos
i<-matrix(c(1,1,1,1,1), 5);i
```

```
     [,1]
[1,]    1
[2,]    1
[3,]    1
[4,]    1
[5,]    1
```

```
#inversa de s
Sinv<-solve(s) %*% i; Sinv
```

```
                    [,1]
AAPL.Close  0.1279400917
MSFT.Close  0.2131991942
NVDA.Close -0.0618950523
AMZN.Close  0.0795316097
META.Close -0.0001091876
```

```r
#calculos para el portafolio optimo
Su<-solve(s) %*% mu; Su
```

```
                    [,1]
AAPL.Close  0.0182337953
MSFT.Close -0.0003328264
NVDA.Close  0.0194274381
AMZN.Close -0.0152974980
META.Close -0.0001234935
```

```r
A<-c(t(i) %*% solve(s) %*% i);A
```

```
[1] 0.3586667
```

```r
B<-c(t(i) %*% Su);B
```

```
[1] 0.02190742
```

```r
C<-c(t(mu) %*% Su);C
```

```
[1] 0.005242642
```

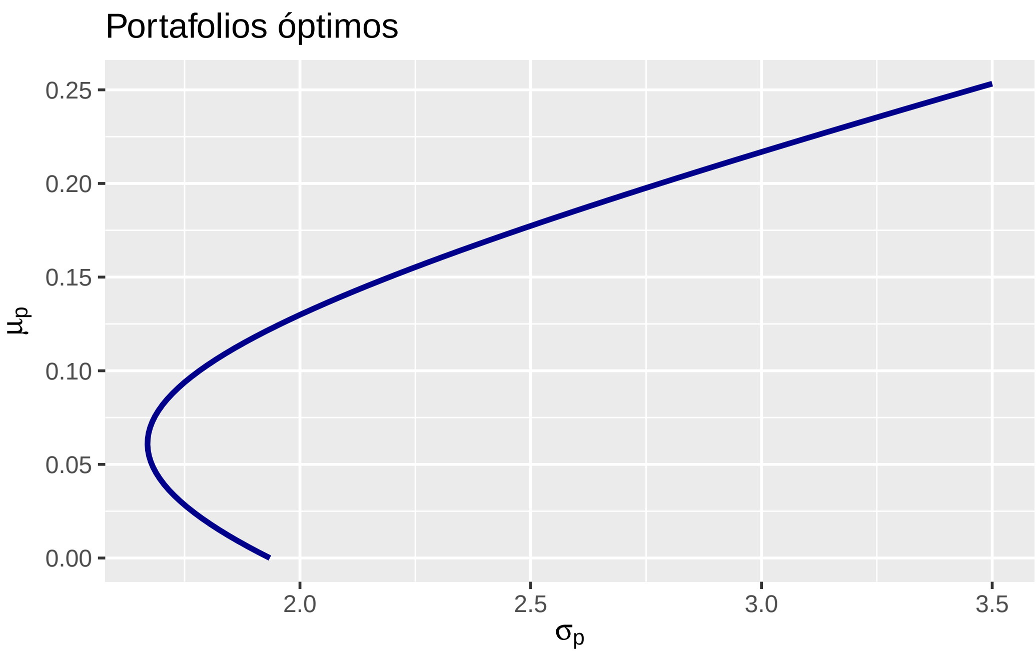
```r
D<- A*C-B^2
```

### Portafolios Óptimos

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v lubridate 1.9.4     v tibble    3.2.1
v purrr     1.0.4     v tidyr     1.3.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::first()  masks xts::first()
x dplyr::lag()    masks stats::lag()
x dplyr::last()   masks xts::last()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
x<-seq(0, 3.5, length=100)
y<-seq(0,0.3, length=100)

df<-expand.grid(x=x, y=y) %>%
  mutate(z=D*x^2-A*y^2+2*B*y-C)
#frontera eficiente
p<-ggplot(df, aes(x=x, y=y, z=z))+
    geom_contour(bins=0, lwd=1, colour="blue4")+
    labs(x=expression(sigma[p]), y=expression(mu[p]))

p+ggtitle("Portafolios óptimos")
```



### Portafolio con retorno 0.05 y mínimo global

```
mu.bar<-0.05

W<-((C-mu.bar*B)/D)*Sinv+((mu.bar*A-B)/D)*Su; W
```

```
                     [,1]
AAPL.Close  3.271431e-01
MSFT.Close  6.323201e-01
NVDA.Close -2.384287e-01
AMZN.Close  2.789385e-01
META.Close  2.709391e-05
```

```
#Varianza
a<-sqrt((mu.bar^2*A-2*mu.bar*B+C)/D); a
```

```
[1] 1.679151
```

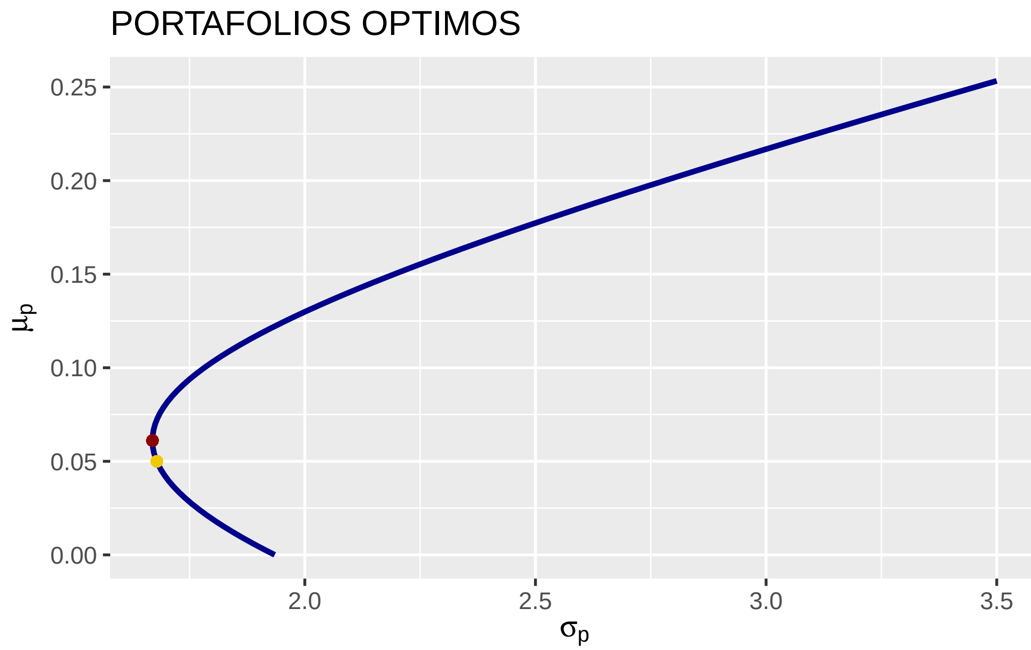
```
#minimo global
w.g<-Sinv/A; w.g
```

```
                     [,1]
AAPL.Close  0.3567103037
MSFT.Close  0.5944215635
NVDA.Close -0.1725698538
AMZN.Close  0.2217424130
META.Close -0.0003044264
```

```
#Riesgo mínimo
s.g<-1/sqrt(A);s.g
```

```
[1] 1.669762
```

```
mu.g<-t(w.g) %*% mu
p+geom_point(aes(x=a, y=mu.bar), colour="gold2")+geom_point(aes(x=c(s.g), y=c(mu.g)), colour=
```




**Portafolio con retorno 0.1 y mínimo global**

```
mu.bar<-0.1

W<-((C-mu.bar*B)/D)*Sinv+((mu.bar*A-B)/D)*Su; W
```

```
                     [,1]
AAPL.Close  0.460567380
MSFT.Close  0.461300124
NVDA.Close  0.058764226
AMZN.Close  0.020837184
META.Close -0.001468915
```

```
a<-sqrt((mu.bar^2*A-2*mu.bar*B+C)/D); a
```

```
[1] 1.782148
```

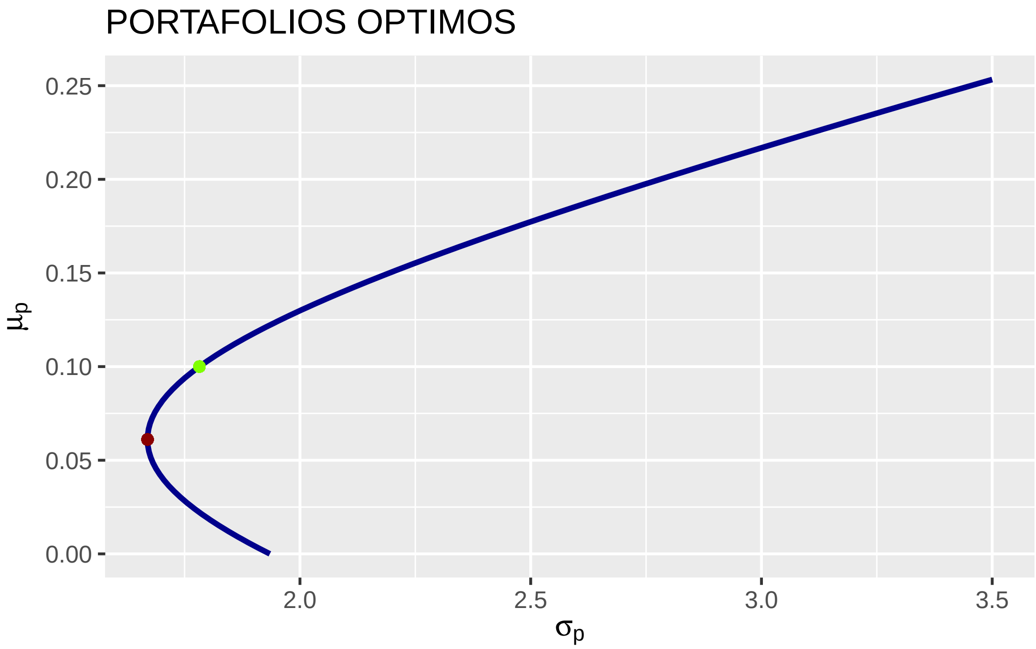
```
w.g<-Sinv/A; w.g
```

```
                    [,1]
AAPL.Close  0.3567103037
MSFT.Close  0.5944215635
NVDA.Close -0.1725698538
AMZN.Close  0.2217424130
META.Close -0.0003044264
```

```
s.g<-1/sqrt(A);s.g
```

```
[1] 1.669762
```

```
mu.g<-t(w.g) %*% mu

p+geom_point(aes(x=a, y=mu.bar), colour="chartreuse1")+geom_point(aes(x=c(s.g), y=c(mu.g)), c
```




**Portafolio con retorno 0.15 y mínimo global**

```
mu.bar<-0.15

W<-((C-mu.bar*B)/D)*Sinv+((mu.bar*A-B)/D)*Su; W
```

```
                       [,1]
AAPL.Close  0.593991705
MSFT.Close  0.290280115
NVDA.Close  0.355957201
AMZN.Close -0.237264096
META.Close -0.002964924
```

```
a<-sqrt((mu.bar^2*A-2*mu.bar*B+C)/D); a
```

```
[1] 2.193882
```

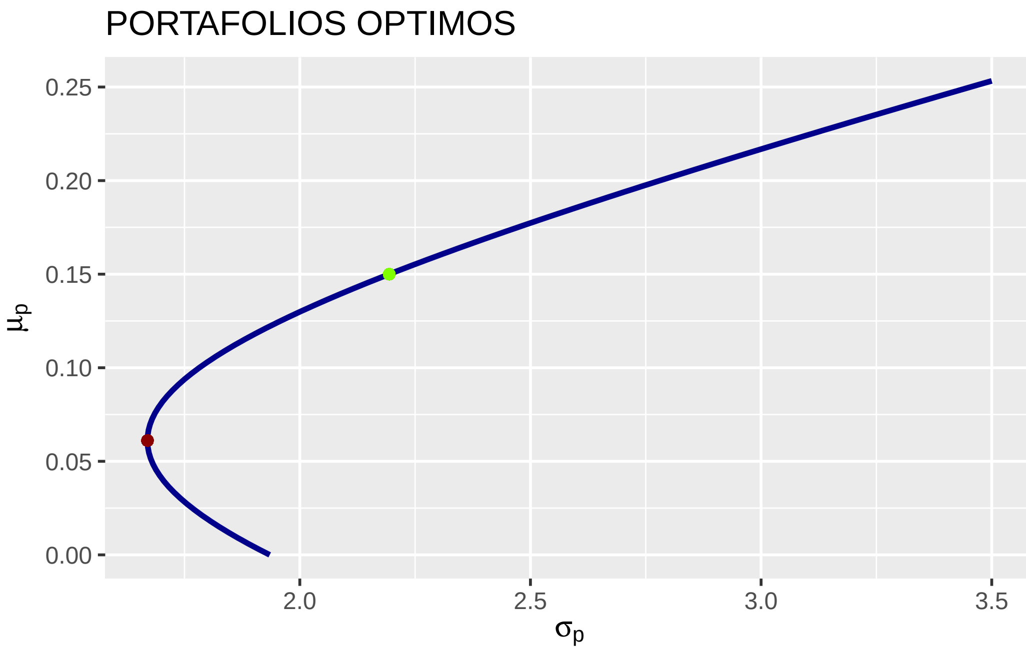
```
w.g<-Sinv/A; w.g
```

```
                        [,1]
AAPL.Close  0.3567103037
MSFT.Close  0.5944215635
NVDA.Close -0.1725698538
AMZN.Close  0.2217424130
META.Close -0.0003044264
```

```
s.g<-1/sqrt(A);s.g
```

```
[1] 1.669762
```

```
mu.g<-t(w.g) %*% mu

p+geom_point(aes(x=a, y=mu.bar), colour="chartreuse1")+geom_point(aes(x=c(s.g), y=c(mu.g)), c
```


PORTAFOLIOS OPTIMOS
0.25
0.20
0.15
0.10
0.05
0.00
$\mu_p$
2.0
2.5
3.0
3.5
$\sigma_p$